Ram Moment and Polynomial Certificates for Four-State Potts Reconstruction

Technical Report M437
An audit and conditional extension of the pinned October 5, 2026 manuscript

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Abstract

The assigned manuscript asserts an exact Kesten–Stigum threshold for the ferromagnetic four-state Potts broadcast model on regular and Poisson Galton–Watson trees. Its essential finite obligation is a pair of rational inequalities on a product of posterior simplices. This report traces the actual source from the Bayesian change of measure through its cone constraint, strict cubic information loss, polynomial lower bounds, subdivision coverage and implementation bounds. The source's 4479- and 1935-cell traversals were *not* replayed here; the global theorem is therefore not independently certified by this report.

The direct Ram contribution is an exact finite moment realization of the tensor-simplicial Bernstein tests, with explicit Gram matrices, inversion domains, transformed subdivision maps and a failure example for testing moment signs instead of polynomial coefficients. Independently executed rational calculations reproduce all 904 scalar coefficients' signs and four stated minima. Their normalized coefficients additionally prove the sharp scalar bound $H \geq -3/25$. Local proofs handle vanishing denominators and the uniform-message zero set. Conditional on the source's remaining pairwise inequalities, an all-merges argument improves the cubic constant, extends the nonreconstruction implication to finite-mean offspring and observed independent random edge strengths, and gives explicit finite-depth bounds. A further conditional argument yields almost-sure quenched rates strictly below the critical annealed exponent and survival-conditioned bounds. Finite tests, source claims, proved local facts and conditional consequences remain separate throughout.

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1 Evidence boundary and pinned problem

1.1 What has and has not been established here

The public source is OpenAI, *The Reconstruction Threshold for the Ferromagnetic Four-State Potts Model*, dated October 5, 2026, in the snapshot

`adc7f1241b42e322a6451854ab7e4b4c146bf78a`.

Its source-local directory is identified in the provenance section. All seven substantive section files, main file, references and the actual C++ text were read. The local package contains the C++ listing as `certificate-source.txt`; a claim that an appendix prints a program is not treated as a substitute for reading that program. No source or private original was changed.

Status	Mathematical content
Source assertion	Exact threshold in the two assigned tree families; completed full subdivision counts; global pairwise inequalities.
Independently proved	Bayes density correspondence; boundary continuity; midpoint coverage; finite Ram coordinate transfer; all-merges implication and rate derivation under the named pairwise premises.
Exact local certificate	Scalar coefficient signs, their normalizations, sharp H lower bound, positive G lower bound and a degree-one moment-inversion example.
Finite diagnostics	441 rational posterior pairs and 52,416 coefficient-multiplier identities; these are not the full product-simplex verification.
Parent evidence	A separately supplied proof note and 3219-assertion exact-check receipt were read and attributed. They do not supply the missing full traversal either.
Unresolved here	Independent execution of both full C++ modes, all accepted terminal coefficient arrays and an execution-backed full coverage receipt.

There is no claimed new solution of the Potts reconstruction problem, no claim of historical novelty for the extensions, and no evidence of a computational speedup from Ram coordinates. The report's useful advance is a precise certificate interface and some checkable strengthened local statements.

1.2 The observation model and every parameter

Let $[4] = \{1, 2, 3, 4\}$ and let the root spin $I = \sigma_\rho$ be uniform. Each edge independently uses

$$P_\lambda(j \mid i) = \lambda \mathbf{1}_{\{i=j\}} + (1-\lambda)/4, \qquad i, j \in [4], \quad 0 \le \lambda \le 1. \tag{1.1}$$

Thus λ is the channel's nonconstant eigenvalue, not the probability of keeping the parent color. That probability is $(1+3\lambda)/4$. The restriction $\lambda \geq 0$ is essential.

The source considers: (i) every integer $d \geq 2$, with exactly d children at each vertex; and (ii) every real $d > 0$, with independent Poisson(d) offspring. Here d is the number of children or its mean, not undirected vertex degree. The tree is independent of root color and edge randomness. At depth n, L_n is the set of vertices at distance n. The observer sees the entire unlabelled rooted tree and its level-n spins, with vertices distinguished by their positions in the observed tree, not by a hidden ordering. Define

$$a_n = \mathbb{E}\left[\frac{1}{2}\sum_{i=1}^{4}\left|\mathbb{P}(I = i \mid T, \sigma_{L_n}) - \frac{1}{4}\right|\right]. \tag{1.2}$$

In the random-tree model the outer expectation includes both tree and broadcast randomness, *without conditioning on survival*. An empty observed level carries no root-spin information. Nonreconstruction means $a_n \to 0$.

The source asserts nonreconstruction whenever $d\lambda^2 \leq 1$, including equality. It combines this with the known above-threshold reconstruction direction for $d\lambda^2 > 1$ in the two specified families. We audit the new nonreconstruction mechanism. The above-threshold argument is not re-proved as a general branching-process theorem here. In particular, results later obtained for arbitrary offspring or random edge strengths are nonreconstruction implications, not automatic two-sided threshold theorems.

1.3 Current public primary context

The October 7, 2026 search used only public mathematical titles and topics. Sly's primary preprint establishes sharpness for three colors at large regular degree and nonsharpness for at least five colors [2]. Mossel–Sly–Sohn's revised primary text, Theorem 1.12, includes equality and treats three and four colors for sufficiently large mean degree under its offspring hypotheses; its Theorem 1.13 distinguishes the four-color antiferromagnetic behavior [3]. These are not all-degree ferromagnetic four-color results. The source's proposed contribution is removal of the large-degree restriction for regular and Poisson offspring.

Gu's June 2026 primary abstract describes a computer-assisted population-dynamics proof for symmetric binary hypertree models [4]. It is relevant as a rigorous computational comparator, not a proof for this four-color model. Leroy's primary paper studies certified polynomial minimization by simplicial Bernstein subdivision [5]; that is established approximation-theory machinery, not a Ram invention. The current public M437 README was found, but the pinned local text, rather than a mutable main-branch page, determines this report's mathematical target. Search discovery is not an exhaustive novelty investigation.

2 Posterior laws and order of expectation

2.1 Coordinates and information

Write $\Delta_4 = \{p \in [0,1]^4 : \sum_i p_i = 1\}$ and $\boldsymbol{u} = (1/4, 1/4, 1/4, 1/4)$. Use

$$L = \begin{pmatrix} 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}, \quad x = Lp, \quad p = \boldsymbol{u} + \tfrac{1}{4}L^{\mathsf{T}}x. \tag{2.1}$$

The domain of x is the tetrahedron $L\Delta_4$, not the entire cube $[-1,1]^3$. Define

$$A = \sum_j x_j^2 = 4\sum_i (p_i - 1/4)^2, \quad B = x_1x_2x_3, \quad C = x_1^2x_2^2 + x_1^2x_3^2 + x_2^2x_3^2. \tag{2.2}$$

Since $L^\mathsf{T}L = 4I - \mathbf{1}\mathbf{1}^\mathsf{T}$, $0 \le A \le 3$, and Cauchy–Schwarz gives

$$\mathrm{TV}(p, \boldsymbol{u}) \le \tfrac{1}{2}\sqrt{A(p)}, \qquad \mathbb{E}\,\mathrm{TV}(p, \boldsymbol{u}) \le \tfrac{1}{2}\sqrt{\mathbb{E}A(p)}. \tag{2.3}$$

The degree of A in x is two. In particular a cubic saving in its mean is a sixth-order information effect, not a cubic polynomial in the original posterior coordinates.

2.2 Marginal law versus combined observation law

For an experiment Y about uniform I, let κ_i be its conditional law and $\kappa = \frac{1}{4}\sum_i \kappa_i$ its marginal. Its posterior satisfies

$$p_i(Y) = \tfrac{1}{4}\frac{\mathrm{d}\kappa_i}{\mathrm{d}\kappa}(Y), \qquad \int p_i \,\mathrm{d}\mu = 1/4,$$

where μ is the pushforward of κ by p. Conversely, any probability law μ on Δ_4 with barycenter $\boldsymbol{u}$ is realized by conditional laws $4p_i\,\mathrm{d}\mu(p)$. Symmetry means invariance of the law under all coordinate permutations; an individual posterior need not have equal coordinates.

If two observations are independent *conditional on the input*, and their marginal posterior laws are μ, ν, set

$$U_i = 4p_iq_i, \quad z = \sum_i U_i, \quad P_i = U_i/z \ (z > 0), \qquad \mathbb{E}_{\mu\star\nu} f = \mathbb{E}_{\mu\otimes\nu}\{z f(P)\}. \tag{2.4}$$

At $z = 0$, set $P = \boldsymbol{u}$ for definiteness. This value is irrelevant after multiplication by z.

Derivation of the density and normalization. Relative to $\kappa \otimes \eta$, the actual joint observation has density $(1/4)\sum_i (4p_i)(4q_i) = 4\sum_i p_iq_i = z$. Bayes' rule divides the ith likelihood contribution by this density and yields P_i. Under the independent marginal law, $\mathbb{E}z = 4\sum_i (\mathbb{E}p_i)(\mathbb{E}q_i) = 1$. Simultaneous color permutation preserves z and permutes P, proving symmetry of the combined law. □

There are two different expectations in (2.4). Factorization of products of functions of p and q is valid under $\mu \otimes \nu$, before tilting by z. Treating p, q as independent under the actual combined observation law would invalidate the cone calculation.

For a continuous invariant function ϕ, define

$$[z\phi(P)](p, q) = \frac{1}{24}\sum_{\pi\in\mathfrak{S}_4} z_\pi \phi(P^\pi), \quad U_i^\pi = 4p_iq_{\pi(i)}, \quad z_\pi = \sum_i U_i^\pi. \tag{2.5}$$

Symmetry of ν lets one replace $z\phi(P)$ by this average inside the product-marginal expectation. It does not make each summand equal pointwise.

2.3 Edges and observed mixing labels

One edge maps a child posterior to

$$C_\lambda(p) = \boldsymbol{u} + \lambda(p - \boldsymbol{u}), \qquad LC_\lambda(p) = \lambda Lp, \quad A(C_\lambda p) = \lambda^2 A(p). \tag{2.6}$$

This follows by conditioning on the child spin and using the symmetric doubly stochastic channel. The marginal observation law is unchanged. An observed root-independent label S with conditional posterior laws μ_s yields the mixture $\int \mu_s \, \mathrm{d}\mathbb{P}_S(s)$. An unobserved label is a different experiment; its posterior is a conditional average, not in general the same mixture of posterior laws.

3 The concrete cone and its dependency chain

3.1 No unspecified Lyapunov function

The source's functions are fixed, not fitted anew in this report:

$1000 \times$ coefficient	1	A	B	A^2	C	AB	A^3	AC	B^2
H	0	0	1000	−100	−150	300	−20	70	−510
F	0	4000	−3000	700	−4000	−800	0	0	0
G	1000	−1610	−3100	773	4000	−3980	0	0	0

(3.1)

Thus $H = B - A^2/10 - 3C/20 + 3AB/10 - A^3/50 + 7AC/100 - 51B^2/100$. The invariants in order have degrees $0, 2, 3, 4, 4, 5, 6, 6, 6$. Define

$$D_0(p,q) = \frac{A(p)A(q)(A(p)+A(q))}{1000}, \qquad \mathcal{C} = \{\mu : \mu \text{ symmetric posterior law}, \ \mathbb{E}_\mu H \geq 0\}.$$

The cone condition is on the *law*. For example $H(1/2, 1/2, 0, 0) = -3/25$; the source does not assert pointwise nonnegativity of H. The fully revealing posterior law assigns mass $1/4$ to each e_i, where $H(e_i) = 13/100$; the uninformative law is $\delta_{\boldsymbol{u}}$, where $H = 0$.

3.2 The exact remaining premises

The independently reproduced scalar certificates imply $F, G \geq 0$ and

$$H(tLp) \geq t^3 H(Lp), \quad p \in \Delta_4, \quad 0 \leq t \leq 1. \tag{3.2}$$

The two remaining source inequalities, required for *every* $p, q \in \Delta_4$, are

$$[zA(P)] \leq A(p) + A(q) - H(p)F(q) - F(p)H(q) - D_0(p,q), \tag{3.3}$$

$$[zH(P)] \geq H(p)G(q) + G(p)H(q). \tag{3.4}$$

In this report, call their conjunction **Premise PC**. Source Sections 5–7 supply the proposed proof by full finite verification. Our sample tests do not establish PC.

Proposition 3.1 (Conditional closure and saving)**.** *Under PC, $\mathcal{C}$ is closed under the edges* (2.6)*, conditionally independent products and observed root-independent mixtures. For $\mu, \nu \in \mathcal{C}$,*

$$\mathbb{E}_{\mu\star\nu} A \leq \mathbb{E}_\mu A + \mathbb{E}_\nu A - \frac{\mathbb{E}_\mu A^2 \mathbb{E}_\nu A + \mathbb{E}_\mu A \mathbb{E}_\nu A^2}{1000}. \tag{3.5}$$

Proof. For edges, take expectation of (3.2). For products, integrate (3.4) under $\mu \otimes \nu$. Equation (2.4) and symmetry identify the left side with the new H expectation. Independence before the tilt makes the right side $\mathbb{E}_\mu H \mathbb{E}_\nu G + \mathbb{E}_\mu G \mathbb{E}_\nu H \geq 0$. Integrating (3.3) in the same measure permits discarding the two nonnegative products involving F and H. The remaining product of powers of A factors exactly to (3.5). Mixture closure is linearity of expectation; symmetry is preserved in all three operations. All integrands are bounded on the compact simplex, so no unproved moment-integrability interchange occurs. □

The source then starts with $m_0 = 3$ and applies this induction at every finite depth. Combining only the first two of k equally informative branches, with $b = \lambda^2 m_n$, gives the source bound

$$m_{n+1} \le d\lambda^2 m_n - \frac{2}{1000}\mathbb{P}(D \ge 2)\lambda^6 m_n^3. \tag{3.6}$$

This is the actual dependency order: scalar signs and scaling; PC; law closure; tree recursion; decay; TV comparison. A polynomial point check cannot be substituted for PC halfway through this chain.

3.3 Whole tree, ordering and extinction

For the induction one can number the children and reveal the ordered truncation $\widehat{T}_{\le n}$ along with its terminal spins. Let $p^{(n)}$ be this posterior and $m_n = \mathbb{E}A(p^{(n)})$. The offspring variables below depth n are independent of the root and of all spins used in that observation, conditional on the truncation. Revealing the rest of the ordered tree therefore leaves the posterior unchanged. Erasing the child ordering is a measurable loss of information. Conditional Jensen and (2.3) give $a_n \le \frac{1}{2}\sqrt{m_n}$ for the source's observation convention.

When $L_n = \varnothing$, the entire spin-independent tree gives posterior $\boldsymbol{u}$. This includes finite extinct trees and is why the unconditioned average requires no separate division by survival probability. Survival conditioning changes the distribution of the tree and invalidates use of an unchanged independent-offspring recursion. Bounds can instead be transferred by conditioning nonnegative quantities, as proved in Section 9.2; this is not a survival-conditioned iid recursion. At Poisson mean one survival has probability zero, so such conditioning is unavailable.

4 Polynomial targets, denominators and boundary zeros

4.1 Ordered simplex reduction

Let $\mathcal{S} = \{p \in \Delta_4 : p_1 \ge p_2 \ge p_3 \ge p_4\}$ and

$$S_i = \frac{1}{i+1}\sum_{v=1}^{i+1} e_v \quad (0 \le i \le 3).$$

With $p_5 = 0$, $X_i = (i+1)(p_{i+1} - p_{i+2})$ are nonnegative, sum to one and satisfy $p = \sum_i X_i S_i$. Thus $\mathcal{S}$ is a genuine three-simplex. Color permutations act on x as coordinate permutations with an even number of sign changes: the exact matrix is $L\Pi L^{\mathsf{T}}/4$. Consequently A, B, C are invariant. The averages (2.5) are invariant under separate permutations of p and q, so $\mathcal{S} \times \mathcal{S}$ suffices, including its equality faces.

Let $\mathcal{I}_r = \{I \in \mathbb{N}_0^4 : |I| = r\}$ and $W_r(X) = (\sum_i X_i)^r$. Homogenization of a monomial X^I of degree at most r multiplies it by $W_{r-|I|}$. In two variable groups, degree (s,t) is elevated to (r,r) using $W_{r-s}(X)W_{r-t}(Y)$. These operations are exact only on the normalized simplices; their coefficient identities must retain the correct degrees.

4.2 The two rational defects

Set

$$K_a = 24\{A(p) + A(q) - H(p)F(q) - F(p)H(q) - D_0(p,q)\},$$
$$K_b = -24\{H(p)G(q) + G(p)H(q)\}.$$

For each permutation use the following exact data:

Row	k	N_π	lower-bound constant m
a	1	$-\sum_{j=1}^{3}(LU^\pi)_j^2$	3
b	5	$z_\pi^6 H(U^\pi/z_\pi)$	1/2

In row b, each degree-s invariant term in H is evaluated at LU^π and multiplied by z_π^{6-s}, so N_π is a polynomial, even when $z_\pi = 0$. Define

$$\mathcal{R} = K + \sum_{\pi\in\mathfrak{S}_4} N_\pi/z_\pi^k. \tag{4.1}$$

The summands are $-z_\pi A(P^\pi)$ for row a and $z_\pi H(P^\pi)$ for row b. Thus $\mathcal{R}_a \geq 0$ and $\mathcal{R}_b \geq 0$ are exactly PC multiplied by 24, with no changed sign convention.

On a cell $p = \sum_i X_i a_i$, $q = \sum_j Y_j b_j$, each U_v^π has coefficient $4a_{iv}b_{j,\pi(v)}$ in bidegree $(1,1)$. The z_π coefficients are nonnegative. N_π has bidegree $(k+1, k+1)$ and K can be represented in $(6,6)$.

Lemma 4.1 (Boundary extension and exact zero set)**.** *Each weighted rational summand extends continuously by zero at $z_\pi = 0$. Furthermore,*

$$\mathcal{R}_a(p, \boldsymbol{u}) = \mathcal{R}_b(p, \boldsymbol{u}) = 0 \quad \textit{for every } p \in \Delta_4, \tag{4.2}$$

and likewise with the arguments interchanged.

Proof. All $U_i^\pi \geq 0$. Hence $z_\pi = 0$ implies $U^\pi = 0$, and for positive z_π the normalized vector lies in Δ_4. The continuous polynomials A, H are bounded there, so the absolute weighted term is at most a constant times z_π, tending to zero. For $q = \boldsymbol{u}$, $z_\pi = 1$ and $P^\pi = p$ for every permutation. Also $A(\boldsymbol{u}) = F(\boldsymbol{u}) = H(\boldsymbol{u}) = 0$ and $G(\boldsymbol{u}) = 1$. Substitution into the two defects cancels them exactly. □

Continuity of P itself is not asserted at a zero denominator, nor required. A global positive lower bound for the pairwise defects is impossible by (4.2). This is a substantive obstruction to a generic "subdivide until all coefficients are strictly positive" termination argument.

4.3 Why direct denominator clearing is possible but costly

On the relative interior of any full-dimensional subsimplex of $\mathcal{S}$, each probability coordinate is positive: an affine nonnegative coordinate vanishing at an interior point would vanish at every vertex, forcing the whole cell into a coordinate face. Thus all $z_\pi > 0$ on the cell-product interior. Multiplication by $\prod_\pi z_\pi^k$ there converts (4.1) to a polynomial without changing sign. With full homogenization, its bidegree can reach $(30, 30)$ in row a and $(126, 126)$ in row b. This naive representation explains why the lower-degree Type T/P route is attractive. Positivity of a denominator product on the interior and continuity are still needed to infer the boundary statement; the cleared polynomial alone can vanish on an entire denominator face without controlling the limiting quotient.

5 Exact sign tests and coverage

5.1 Type T: a global Taylor minorant

For $k = 1$ or 5, let

$$t_k(z) = \sum_{j=0}^{5} \binom{-k}{j} (z-1)^j.$$

The sixth derivative of z^{-k} is $k(k+1)\cdots(k+5)z^{-k-6} > 0$. Taylor's theorem about 1 gives $z^{-k} - t_k(z) \geq 0$ for *all* $z > 0$, not only $|z-1| < 1$. Since $A \leq 3$ and $H \geq -1/2$, in each row $N_\pi + mz_\pi^{k+1} \geq 0$. Therefore

$$T = K + \sum_\pi \{(N_\pi + mz_\pi^{k+1}) t_k(z_\pi) - mz_\pi\} \leq \mathcal{R}. \tag{5.1}$$

The homogeneous degree is $(k+6, k+6)$, namely $(7,7)$ or $(11,11)$. Nonnegative ordinary coefficients imply $T \geq 0$ on the full product simplex; Lemma 4.1 transfers the resulting rational inequality to the boundary. Multiplying the Taylor bound directly by N_π would be wrong because N_π can be negative. The compensating mz_π^{k+1} is essential.

5.2 Type P: a polynomial proposal with an exact remainder

Choose any rational bihomogeneous J_π of bidegree $(3,3)$. The source proposes it using the 20 simplex lattice nodes $I/3$, $I \in \mathcal{I}_3$, in each argument. Its cardinal polynomials are

$$\ell_I(X) = \prod_{i=0}^{3} \binom{3X_i}{I_i}.$$

At $J/3$, this is $\mathbf{1}_{I=J}$. Their homogeneous forms replace every linear factor $3X_i - h$ with $3X_i - h\sum_j X_j$. Interpolation uses 400 node pairs, truncating each rational value toward zero to a multiple of M^{-1}, $M = 2^{32}$. Zero-denominator node values are assigned the continuous value zero.

Interpolation accuracy is not assumed. Instead form

$$O_\pi = W_2(X)W_2(Y)N_\pi - z_\pi^k J_\pi, \qquad Q_\pi = z_\pi^k W_3(X)W_3(Y). \tag{5.2}$$

Both have bidegree $(k+3, k+3)$ and Q_π has nonnegative coefficients. If any negative coefficient O_{IJ} has $Q_{IJ} = 0$, the test must fail. Otherwise choose

$$c_\pi = M^{-1} \max\left(\{0\} \cup \{\lceil -MO_{IJ}/Q_{IJ} \rceil : O_{IJ} < 0\}\right). \tag{5.3}$$

Then $O_\pi + c_\pi Q_\pi$ is coefficientwise nonnegative, and on the interior

$$\frac{N_\pi}{z_\pi^k} = J_\pi - c_\pi + \frac{O_\pi + c_\pi Q_\pi}{z_\pi^k} \geq J_\pi - c_\pi.$$

The final test is coefficientwise nonnegativity, after elevation to $(6,6)$, of $K + \sum_\pi (J_\pi - c_\pi W_3(X)W_3(Y))$. This proves the complete region, not just the interpolation nodes. The rounding direction used to construct J_π does not itself prove a bound; the exact remainder does.

5.3 Midpoint coverage and its coefficient action

Replacing vertex a_i by $(a_i + a_j)/2$, $i \neq j$, gives old barycentric coordinates $X_i = X'_i/2$, $X_j = X'_j + X'_i/2$. Consequently an old coefficient at $I \in \mathcal{I}_r$ contributes to

$$I + t\varepsilon_i - t\varepsilon_j \quad (0 \le t \le I_j), \qquad 2^{-I_i - t}\binom{I_j}{t} \tag{5.4}$$

times that coefficient. The same acts in Y for subdivision of the second factor.

Lemma 5.1 (Coverage without a sampling argument)**.** *The two children replacing i and j, respectively, cover the parent simplex and remain affinely independent. A finite recursively accepted binary tree of product cells therefore covers the initial product domain.*

Proof. The first child is exactly $X_i \le X_j$: set $X'_i = 2X_i$, $X'_j = X_j - X_i$. The second is $X_j \le X_i$. Both inequalities include the equality face and together exhaust the simplex. Each substitution has determinant $1/2$ in barycentric coordinates and so preserves nondegeneracy. Taking a product with the unchanged second factor preserves the cover, and finite induction completes the argument. □

The source tries T first, then P only when neither vertex list contains $\boldsymbol{u}$. Failure of the correction denominator assertion aborts. If neither test succeeds at depth 24, it reports failure. Otherwise it maximizes lexicographically (w, y, i, j), where y selects the factor, $i > j$, and w is squared edge length multiplied by 16 when an endpoint is uniform. It checks the child replacing j first, then the child replacing i. These ordering rules matter for reproducing counts; they are not necessary for the abstract coverage lemma.

5.4 Source counts and the unexecuted obligation

Row	Visited	Maximum depth	T leaves	P leaves	Unresolved
a	4479	19	166	2074	0
b	1935	20	97	871	0

These are **source-reported counts**, not output from our execution. They satisfy the necessary full-binary-tree consistency check $N = 2L - 1$: $4479 = 2(2240) - 1$ and $1935 = 2(968) - 1$. That arithmetic does not verify a single terminal coefficient array.

A complete independent replay should retain both-mode exit codes with assertions enabled, exact accepted-cell signs, exact vertex genealogy, the root cell, both children at every split, nondegeneracy and the handling of all apparent denominator singularities. Merely validating a recorded log against another summary is not the same operation. No suitable C++ compiler was reported on PATH in the parent environment, and this report did not install one or run either complete mode.

6 Implementation audit of the actual source

The read C++ program uses `boost::multiprecision::cpp_int` for coefficient numerators and common denominators, and fixed-width integers for combinatorics, indexing and scaled vertices. The key correspondence is as follows.

Routine	Mathematical obligation observed in source
`init, coef, ones`	Descending multi-indices, multinomials and $W_r(X)W_r(Y)$.
`operator*, lift, +`	Ordinary coefficient convolution and common-denominator degree elevation, not pointwise multiplication of coefficient entries.
`inv, dot`	The nine exact invariants and division by 1000; use of z rather than homogeneous one supplies numerator homogenization.
`tensor, make`	Separate X/Y factors recover their univariate coefficients from a pure-power column or row; the factor 24 and strict D_0 are present.
`sub`	Formula (5.4) with 2^r cleared into the common denominator.
`ev, inverse, act`	Exact node powers and cardinal coefficients; evaluation divides by 3^{2r} and two-sided interpolation by 960^2.
`testP`	Truncation at 2^{-32}; construction of O, Q; assertion $Q_{IJ} > 0$ for negative O_{IJ}; integer ceiling correction; final sign test.
`run`	T then eligible P; depth-cap failure; exact weighted edge tie-break; both recursive children.

Common denominators start positive; addition uses positive common factors, multiplication multiplies them, and explicit divisors are positive. Thus sign tests on numerators have the intended meaning. Assertions must not be disabled. This is an inspection of the correspondence, not a formally verified compiler or memory-safety proof.

The source's relevant size bounds can be checked analytically. With $\kappa = 12 \cdot 2^{24}$, vertices remain in $[0, \kappa]$ and midpoint coordinates are integers through depth 24. The fourth coordinate equals $\kappa/4$ only at the uniform vertex in the ordered simplex. The largest working bidegree is 11, giving $\binom{14}{3} = 364$ indices and 132,496 entries. The depth cap admits at most $2^{25} - 1$ visited cells. The largest initialized factorial is $19! < 2^{63}$ and the square of the largest degree-11 multinomial is $92400^2 < 2^{63}$. A conservative weighted-distance bound is

$$64\kappa^2 = 2\,594\,073\,385\,365\,405\,696 < 2^{63}.$$

The recurrence multiplier with denominator 2^r cleared is $2^{r-I_i-t}\binom{I_j}{t}$; its next-step integer numerator is bounded by $11 \cdot 2^{11}\binom{11}{5} < 2^{24}$. The recurrence's division is exact before the last unused zero term. Cardinal construction is bounded by $960 \cdot 8^3 < 2^{19}$; evaluation entries are at most 3^6. These bounds support the stated fixed-width ranges. They do not demonstrate that an unrun traversal reaches successful leaves.

7 Direct Ram correspondence, not a renamed algorithm

7.1 The private passages actually used

Three confidential local materials supplied concrete constraints on the reformulation. TR250's "What a Ram representation means" defines finite moment coordinates $m = Ec$, synthesis $c = E^{-1}m$, and warns that finite invertibility does not give uniform stability. Its concrete polynomial construction uses positive weighted Gram moments on a specified polynomial space. Volume 16's clearance chapter proves exact power-to-Bernstein conversion and convex-hull enclosures, with an explicit warning that node satisfaction is not a continuous certificate and depth exhaustion is inconclusive. TR-core-pat-16 distinguishes exact polynomial moment contraction from posterior adequacy; its

PSD, remainder and rank-loss sections show why positivity and finite point inverses alone are inadequate [6, 7, 8].

None of their PDE, trajectory, neural-inference or inverse-estimator theorems directly establishes the tree threshold. We use the finite polynomial moment realization, exact enclosure logic and explicit remainder discipline. Their private contents were read locally only and were not sent as public search terms or uploads.

7.2 Concrete tensor-simplex moment space

On the reference tetrahedral simplex $X \in \Delta_4$, let

$$B_I^r(X) = \binom{r}{I} X^I, \qquad I \in \mathcal{I}_r.$$

These form a basis of the polynomials of degree at most r restricted to $\sum X_i = 1$. There are $n_r = \binom{r+3}{3}$ basis elements; there is no redundancy among the fixed homogeneous-degree basis restricted to the simplex. For degree (r, s) on the product, the basis is $B_I^r(X)B_J^s(Y)$. In particular, source ordinary coefficients c_{IJ} correspond to

$$\beta_{IJ} = \frac{c_{IJ}}{\binom{r}{I}\binom{s}{J}}. \tag{7.1}$$

The sign is unchanged, but *the magnitude is not.* An ordinary coefficient minimum is not in general a lower bound for the polynomial; a Bernstein coefficient minimum is, because the basis is a nonnegative partition of unity.

Let ω be normalized uniform probability measure on the reference simplex, so $\int 1 \, \mathrm{d}\omega = 1$. Define polynomial Ram moments

$$m_{IJ}(R) = \int_{\Delta_4^2} B_I^r(X)B_J^s(Y)R(X, Y) \, \mathrm{d}\omega(X) \, \mathrm{d}\omega(Y). \tag{7.2}$$

These are moments of a *certificate polynomial against a reference measure.* They are not moments of the random posterior law μ_n. Conflating those two measures would change the problem.

The one-factor Gram matrix is the explicit rational matrix

$$E_{IK}^{(r)} = \int B_I^r B_K^r \, \mathrm{d}\omega = \frac{6(r!)^2}{(2r+3)!} \prod_{h=0}^{3} \frac{(I_h + K_h)!}{I_h! K_h!}. \tag{7.3}$$

This uses $\int X^\alpha \, \mathrm{d}\omega = 6 \prod_h \alpha_h! / (|\alpha| + 3)!$. In a fixed vectorization order, $m = (E^{(r)} \otimes E^{(s)})\beta$.

Proposition 7.1 (Exact Ram realization of the sign test)**.** *For every $r, s \geq 0$, $E^{(r)}$ and $E^{(r)} \otimes E^{(s)}$ are positive definite and rationally invertible. The sufficient coefficient certificate is exactly*

$$(E^{(r)} \otimes E^{(s)})^{-1} m(R) \geq 0 \quad \textit{entrywise.} \tag{7.4}$$

When (7.4) *holds, $R(X, Y) \geq 0$ on the entire product simplex.*

Proof. For nonzero coefficients v, the polynomial $\sum_I v_I B_I^r$ is not identically zero and has strictly positive squared integral under a measure positive on every relatively open set. Thus $v^\mathsf{T} E^{(r)} v > 0$. The product basis has the analogous property, or one may use tensor products of positive eigenvalues. Formula (7.3) is rational, so exact Gaussian elimination gives a rational inverse. It reconstructs precisely the Bernstein coefficient vector, whose nonnegative basis weights sum to one. Their contraction is therefore nonnegative everywhere. □

This proposition identifies the specific Ram space, functionals, normalization, operator and cone. It transfers the established Bernstein certificate. It does not supply a new positivity theorem for arbitrary learned atoms or guarantee a well-conditioned high-degree matrix.

7.3 A small complete example and a necessary negative control

For $r = 1$, in the vertex order X_0, X_1, X_2, X_3,

$$E^{(1)} = \frac{1}{20}(I + \mathbf{1}\mathbf{1}^{\mathsf{T}}), \qquad (E^{(1)})^{-1} = 20I - 4\mathbf{1}\mathbf{1}^{\mathsf{T}}. \tag{7.5}$$

The constant-direction eigenvalue is $1/4$ and the other three are $1/20$, so the Euclidean condition number is 5; the two-factor matrix has condition number 25. The independent script verifies the inverse exactly.

Take $R(X) = X_0 - 1/8$. Its Bernstein coefficients are $(7/8, -1/8, -1/8, -1/8)$ and its moments are

$$m(R) = (11/160, 3/160, 3/160, 3/160).$$

Every moment is positive, yet $R = -1/8$ on $X_0 = 0$. Testing $m \geq 0$ instead of (7.4) is therefore demonstrably unsound even in degree one. Similarly, a PSD posterior covariance matrix only validates certain second-moment identities. It cannot replace the support-realizable law or the rational pairwise inequalities.

7.4 Subdivision and posterior expectations in Ram coordinates

Let D be the positive diagonal conversion from Bernstein to ordinary coefficients, so $c = D\beta$, and let S be an exact ordinary coefficient restriction matrix from (5.4). At fixed degree the correct moment restriction is

$$m' = \mathsf{E}D^{-1}SD\mathsf{E}^{-1}m, \qquad \mathsf{E} = E^{(r)} \otimes E^{(s)}. \tag{7.6}$$

Here each child uses normalized reference-simplex measure in its own barycentric coordinates. Hence no omitted physical-volume determinant appears; a physical-volume moment convention would need that determinant. Equation (7.6) follows by synthesis, substitution and reanalysis in that order. Multiplication can likewise be implemented by exact synthesis, ordinary coefficient convolution, degree conversion and moment analysis. A practical implementation should usually retain the sparse coefficient representation rather than repeatedly invert dense Gram matrices.

There is a second, different moment use for a posterior law. For the nine invariants $\Phi = (1, A, B, A^2, C, AB, A^3, AC, B^2)$ set $v_j(\mu) = \mathbb{E}_\mu \Phi_j$. Then $\mathbb{E}H = h^{\mathsf{T}}v/1000$, $\mathbb{E}F = f^{\mathsf{T}}v/1000$, and similarly for G. A channel step is the exact diagonal map $v_j \mapsto \lambda^{e_j} v_j$, with degree exponents given above. Under the independent product of marginal laws, $\mathbb{E}[\Phi_i(p)\Phi_j(q)] = v_i(\mu)v_j(\nu)$. The actual product output, however, is rational after Bayes normalization. Its nine moments are not asserted to close as a finite polynomial recurrence. PC supplies inequalities sufficient for the tree argument, not an exact finite-dimensional replacement of the posterior distribution.

8 Independent scalar certificates and a sharper lower bound

8.1 Executed coefficient construction

The ordered coordinate forms are

$$x = (X_0 + X_1 + X_2/3,\ X_0 + X_2/3,\ X_0 - X_2/3).$$

The independent Python program builds each invariant with dictionaries of rational monomial coefficients and exact convolution. A degree-e_j invariant is multiplied by W_{r-e_j} to reach degree r. It enumerates *all* $\mathcal{I}_r$, including absent coefficients as zero. The complete scalar coefficient arrays are saved in the audit package, not inferred from sampled evaluations.

Polynomial	Degree	Count	Ordinary minimum	Bernstein minimum
$H + 1/2$	6	84	19/50	19/50
F	5	56	0	0
G	14	680	87/1000	613/71500
$(3 - \mathcal{E})H$	6	84	0	0

Here $\mathcal{E} = \sum_j x_j \partial_{x_j}$. The zero counts in these four rows are $0, 4, 0, 20$, respectively. The ordinary minima reproduce the source's table; the normalized minima are independently computed additions. In particular, 87/1000 is *not* asserted as a pointwise lower bound for G.

Proposition 8.1 (Exact local strengthening)**.** *For every $p \in \Delta_4$,*

$$H(p) \geq -3/25, \qquad G(p) \geq 613/71500 > 0. \tag{8.1}$$

The first lower bound is sharp, attained at $(1/2, 1/2, 0, 0)$ and its permutations. The second is a certified lower bound, not a claimed exact minimum.

Proof. Divide every homogeneous coefficient by its positive multinomial factor. The resulting Bernstein basis sums to one, so the first row gives $H + 1/2 \geq 19/50$, namely $H \geq -3/25$. The third gives the displayed G bound. Permutation invariance extends the ordered-simplex result. At the stated vector, $x = (1, 0, 0)$, so $A = 1$, $B = C = 0$ and $H = -1/10 - 1/50 = -3/25$. □

This is a computer-assisted finite scalar proof, with exact arrays and reproducing code. It is not dependent on PC or on the unexecuted C++ traversal.

The same coefficient check gives $(3 - \mathcal{E})H \geq 0$. Since $tLp \in L\Delta_4$ for $0 \leq t \leq 1$,

$$\frac{\mathrm{d}}{\mathrm{d}t}\{t^{-3}H(tLp)\} = t^{-4}(\mathcal{E} - 3)H(tLp) \leq 0 \quad (t > 0).$$

Comparison with $t = 1$ proves (3.2); $t = 0$ follows from $H(0) = 0$.

8.2 A possible improved Type T test, with its limitation

Proposition 8.1 permits $m = 3/25$ in row b of (5.1), in place of $1/2$. For fixed N, z, k, differentiating the expression with respect to m gives

$$z^{k+1} t_k(z) - z = -z^{k+1}(z^{-k} - t_k(z)) \leq 0.$$

Thus the smaller valid m yields a pointwise stronger polynomial lower bound. Its coefficient signs need not all improve under this change, because pointwise nonnegativity is not equivalent to coefficientwise nonnegativity at a fixed degree. No revised subdivision counts or runtime reduction are claimed. Replaying both traversals with this modified target is future work, and must not be confused with checking the unchanged source program.

9 Conditional all-merges extension and explicit rates

The following extension was suggested in the parent proof note and is proved here with its measure convention intact. It assumes PC, not merely the scalar checks or sampled pairs.

Lemma 9.1 (A cubic saving at every merge)**.** *Let k observations be independent conditional on a uniform four-state input, with identical symmetric marginal posterior laws in $\mathcal{C}$ and mean information b. Let a_j be the actual mean A after combining the first j. Under PC,*

$$a_k \le kb - \frac{2}{1000}(k-1)_+ b^3, \qquad a_0 = 0. \tag{9.1}$$

Proof. Under the actual observation law, the posterior based on the first observation is the conditional expectation of the posterior based on all first j observations. Convexity of A gives $a_j \ge b$ (indeed a_j is nondecreasing). Product closure keeps the combined law in $\mathcal{C}$. The aggregate of the first j observations and observation $j+1$ remain conditionally independent given the input. Apply (3.5) to their independent *marginal* posterior laws and Jensen to the two squared information terms:

$$a_{j+1} \le a_j + b - \frac{a_j^2 b + a_j b^2}{1000} \le a_j + b - \frac{2b^3}{1000}.$$

Summing $j = 1, \dots, k-1$ gives the result. The cases $k = 0, 1$ are exact. The data-processing inequality and the factorization step use different but explicitly specified measures, so no unconditional branch independence is presumed. □

Theorem 9.2 (Finite-mean offspring, conditional on PC)**.** *Let offspring counts be iid nonnegative integers D, independent of root, edges and descendant randomness, with finite mean d. Retain the observed-tree, unconditioned convention. Define*

$$c = \mathbb{E}(D-1)_+ = d - 1 + \mathbb{P}(D = 0), \quad \rho = d\lambda^2, \quad \eta = 2c\lambda^6/1000. \tag{9.2}$$

Then under PC the finite-depth information satisfies

$$0 \le m_{n+1} \le \rho m_n - \eta m_n^3, \qquad m_0 = 3. \tag{9.3}$$

If $\rho \le 1$ and $c > 0$, it tends to zero. If $c = 0$ and $\rho < 1$, it also tends to zero. The case $c = 0, \rho = 1$ is the noiseless infinite line and reconstructs.

Proof. Finite mean implies finite offspring almost surely, hence each finite-depth tree is finite almost surely. Conditional on $D = k$, the k branch laws are identical edge-pushforwards with mean $b = \lambda^2 m_n$. Lemma 9.1 gives $kb - 2(k-1)_+ b^3/1000$. Average over the observed independent count. No second or third offspring moment is needed, because $\mathbb{E}(D-1)_+ \le \mathbb{E}D$. This proves the recursion. For $c > 0$ and $\lambda > 0$, $m_{n+1} \le m_n - \eta m_n^3$ forces the nonnegative decreasing limit to vanish. At $\lambda = 0$, $m_1 = 0$. If $c = 0$, $D \le 1$ almost surely and the geometric bound follows. Finally $d\lambda^2 = 1$, $d \le 1$, $\lambda \le 1$ imply $D = 1$ almost surely and $\lambda = 1$ in the exceptional case. □

The improvement over the source is $c \geq \mathbb{P}(D \geq 2)$, strict when $\mathbb{P}(D \geq 3) > 0$. For regular offspring, $c = d - 1$ rather than 1. For Poisson offspring, $c = d - 1 + e^{-d}$ rather than $1 - (1 + d)e^{-d}$. This is a conditional extension of an existing certificate, not a novelty claim about general Kesten–Stigum theory.

Corollary 9.3 (Explicit finite-depth bounds)**.** *Under Theorem 9.2, if* $0 < \rho < 1$ *and* $\eta > 0$*,*

$$m_n \leq \rho^n \left[\frac{1}{9} + \frac{2\eta}{\rho} \frac{1 - \rho^{2n}}{1 - \rho^2} \right]^{-1/2}. \tag{9.4}$$

At $\rho = 1$*,* $m_n \leq (1/9 + 2\eta n)^{-1/2}$ *and*

$$a_n \leq \tfrac{1}{2}(1/9 + 2\eta n)^{-1/4}. \tag{9.5}$$

For any $\rho \leq 1$*, the weaker critical-shaped bound holds when* $\eta > 0$*. For* $\eta = 0$*,* $m_n \leq 3\rho^n$*. For* $\rho = 0$*,* $m_1 = 0$*.*

Proof. If either of two consecutive values vanishes, subsequent bounds are immediate. Otherwise $v = \eta m_n^2/\rho < 1$ and $(1 - v)^{-2} \geq 1 + 2v$ yield

$$\frac{1}{m_{n+1}^2} \geq \frac{\rho^{-2}}{m_n^2} + 2\eta\rho^{-3}.$$

Multiplying by ρ^{2n+2} and summing gives (9.4), using $m_0^{-2} = 1/9$. At $\rho = 1$ the increments are constant. Replacing ρ by one in the upper recursion gives the weaker bound, and (2.3) gives (9.5). The geometric and zero cases follow directly from (9.3). □

These constants are not optimal. In a critical regular tree $\lambda = d^{-1/2}$, $\eta = (d - 1)/(500d^3)$. A sufficient depth for $a_n \leq \epsilon$ from the critical estimate is

$$n \geq \max\left\{0, \frac{(2\epsilon)^{-4} - 1/9}{2\eta}\right\}.$$

The polynomial rate describes a certified upper bound, not a sharp critical exponent.

9.1 Observed iid random edge strengths

Let each edge have a strength $\Lambda \in [0, 1]$, iid and independent of tree, root and all other edge/subtree randomness. Reveal these strengths along with the tree. Put $q = \mathbb{E}\Lambda^2$. Each edge experiment is an observed independent mixture of channel experiments, so

$$b = qm_n, \qquad \mathbb{E}H(\Lambda Lp) \geq \mathbb{E}\Lambda^3 \mathbb{E}H(Lp) \geq 0.$$

Therefore the same proofs apply with

$$\rho = dq, \qquad \eta = 2cq^3/1000. \tag{9.6}$$

The full environment beyond the truncation is independent of the root conditional on the observed truncation; it adds no information. Hiding independent edge labels can only reduce the expected advantage, so the nonreconstruction upper bound also applies to that less informative experiment. No correlated environment result, spin-dependent offspring model, or above-threshold random-environment theorem is asserted. If $c = 0$ and $\rho = 1$, then $D = 1$ and $\Lambda = 1$ almost surely, the same exception as before.

9.2 Quenched and survival-conditioned consequences

The next consequence was proposed in a subsequent parent derivation and is proved independently below. Let $\mathcal{T}$ denote the observed spin-independent environment: tree and, in the random-channel model, the edge strengths. For a fixed environment, define

$$v_n(\mathcal{T}) = \mathrm{TV}(\mathcal{L}(I, \sigma_{L_n} \mid \mathcal{T}), \mathcal{L}(I \mid \mathcal{T}) \otimes \mathcal{L}(\sigma_{L_n} \mid \mathcal{T})), \tag{9.7}$$

$$m_n(\mathcal{T}) = \mathbb{E}\left[A(\mathbb{P}(I = \cdot \mid \mathcal{T}, \sigma_{L_n})) \mid \mathcal{T}\right]. \tag{9.8}$$

The first quantity is the conditional expected posterior TV advantage. It is not the TV of one realized posterior. Conditional Jensen gives $v_n(\mathcal{T})^2 \le m_n(\mathcal{T})/4$. For an ordered environment, $\mathbb{E}m_n(\mathcal{T}) = m_n$; for the source's unlabelled observation it is at most the ordered m_n used in the proof.

Proposition 9.4 (Almost-sure rates, conditional on PC)**.** *In the critical case $\rho = 1$, $\eta > 0$, for almost every observed environment,*

$$v_n(\mathcal{T}) = O_{\mathcal{T},\beta}(n^{-\beta}) \quad \textit{for every } 0 < \beta < 1/4.$$

If $0 < \rho < 1$, then for almost every environment and every $0 < \gamma < -\frac{1}{2}\log\rho$, one has $v_n(\mathcal{T}) \le e^{-\gamma n}$ for all sufficiently large n, with the starting depth depending on the environment and γ. These statements do not assert the critical endpoint exponent or a uniform deterministic quenched constant.

Proof. Given the environment, $I \to \sigma_{L_n} \to \sigma_{L_{n+1}}$ is a Markov chain. Applying the same transition kernel to the joint law and to its root-independent product law contracts total variation. Thus $v_{n+1}(\mathcal{T}) \le v_n(\mathcal{T})$, including empty levels.

At criticality, (9.5) and the stronger second-moment implication give $\mathbb{E}v_n^2 \le \frac{1}{4}(1/9 + 2\eta n)^{-1/2} \le Cn^{-1/2}/4$ for $n \ge 1$, where $C = (2\eta)^{-1/2}$. For fixed $\beta < 1/4$, Markov's inequality yields

$$\mathbb{P}\{v_{2^k} > 2^{-\beta k}\} \le (C/4)2^{-(1/2-2\beta)k}.$$

The series converges. Borel–Cantelli, which needs no independence between these events, bounds the dyadic subsequence eventually. If $2^k \le n < 2^{k+1}$, monotonicity gives $v_n \le 2^{-\beta k} \le 2^\beta n^{-\beta}$. A countable intersection for rational exponents and then selection of a larger rational exponent proves the statement simultaneously for every real $\beta < 1/4$.

Subcritically, $\mathbb{E}v_n^2 \le 3\rho^n/4$. Hence $\mathbb{P}\{v_n > e^{-\gamma n}\} \le (3/4)(\rho e^{2\gamma})^n$, summable for the stated γ. Borel–Cantelli and the same countable-exponent argument prove the claim. If $\rho = 0$, the advantage vanishes almost surely from depth one, without this logarithmic formula. □

For a survival event S measurable in the environment with $p_S = \mathbb{P}(S) > 0$, root uniformity remains intact because the environment is spin independent. Without claiming iid offspring under this conditioning,

$$\mathbb{E}[v_n^2 \mid S] \le \frac{m_n}{4p_S}, \qquad \mathbb{E}[v_n \mid S] \le \frac{1}{2}\sqrt{\frac{m_n}{p_S}}. \tag{9.9}$$

Indeed, restrict the nonnegative integrand $v_n^2 \le m_n(\mathcal{T})/4$ to S, divide by p_S, and use Jensen for the second bound. Substituting (9.4) or the critical estimate supplies the explicit survival-conditioned annealed bounds. Null environment sets remain null after conditioning on a positive-probability survival event. These consequences do not alter the source's original unconditioned convention or fill the missing PC verification.

10 Finite verification record and error budgets

10.1 What this author actually executed

The audit program uses Python's exact `fractions.Fraction` arithmetic, with assertions enabled. Its full coefficient ledger and summary are separate files. The finite checks were:

1. All 904 scalar coefficient signs, exact ordinary minima and normalized Bernstein minima; these are exhaustive coefficients for the specified four polynomials.
2. All 24 color-permutation actions in the three coordinates, including the even-sign condition.
3. All 441 pairs from the 20 denominator-three simplex nodes plus $\boldsymbol{u}$. Both exact defects were nonnegative, with minima zero. There were 2208 zero-denominator summands, handled by their zero weighted extension. These are *pointwise tests only.*
4. The 21 tested pairs against $\boldsymbol{u}$ had both defects exactly zero. Lemma 4.1, not the finite count, proves the complete zero set used here.
5. All 52,416 midpoint multiplier cases generated by degrees 0 through 11, all multi-indices, all oriented edges and every allowed binomial index. This checks the cleared multiplier identity, not traversal coverage.
6. The degree-one Gram inverse and the positive-moments/negative-polynomial counterexample, in exact arithmetic.

The parent supplied a separate receipt reporting 3219 exact assertions, including 20 actual finite channel-product calculations and 1639 reciprocal-step and 240 closed-form checks. We read that receipt and proof note; the author's independent script does not relabel those tests as its own. Neither suite executes the source's 4479/1935-cell traversal.

10.2 Why finite posterior moments or samples cannot replace PC

An empirical symmetric posterior law can be perfectly normalized, satisfy $\mathbb{E}H \geq 0$, and have a PSD covariance while missing a region where a proposed pairwise inequality fails. The exact Bayes update involves denominators, so even matching finitely many moments does not determine all its rational expectations. The finite moment representation supplies exact integrals for its declared polynomial space, not an arbitrary rational function. Type P is sound precisely because it pays for the actual remainder after interpolation.

For a nonexact Ram implementation, errors must be separated. If $\widehat{m} = m + \delta m$ and $\|\delta m\|_\infty \leq \epsilon$, exact inversion gives

$$\left\|\widehat{\beta} - \beta\right\|_\infty \leq \left\|\mathsf{E}^{-1}\right\|_\infty \epsilon.$$

Coefficient or inverse-matrix errors add their own terms. A positive estimated coefficient margin larger than the complete bound can certify positivity; a zero margin cannot. Near the uniform zero set, a fixed positive error allowance will generally preclude proving nonnegativity by such a test. Rational arithmetic or exact vanishing-factor identities are natural responses. A sampled residual or a PSD matrix is not a replacement error budget.

The rank-loss warning from TR-core-pat-16 is relevant but must not be transferred literally: there $1/A$ can have divergent moments even when $A \neq 0$ almost surely. Here the numerator structure

$N/z^k = z\phi(P)$ is what removes the apparent singularity. Bounding $1/z^k$ in isolation would discard that cancellation and produce an unnecessary divergent or unbounded estimate. The correct object to certify is the complete weighted summand.

11 Advantages, limitations and specific future work

The certificate-oriented Ram formulation has concrete advantages. It keeps normalization and reference measure explicit, supports exact algebraic interchange with existing moment infrastructure, and makes the accepted cone a checkable inverse-image cone rather than an unexplained positivity heuristic. Tensor structure permits factorized transforms. Exact boundary reasoning avoids false failures caused by individually unbounded reciprocals. The all-merges extension extracts a stronger quantitative consequence without changing PC.

There are corresponding costs. Dense Gram inverses can lose sparsity and amplify numerical errors; source coefficient convolution is often a better implementation. A coordinate change does not improve the mathematical approximation space or guarantee fewer cells. Degree-11 bivariate arrays already have 132,496 entries per polynomial, and row-b storage includes many such arrays at recursive cells. Rational numerator sizes and repeated interpolation can dominate runtime. Full positive-margin subdivision theorems do not apply unchanged to the exact uniform-message zero set. Private Ram foundations do not substitute for validating this public source's finite certificate.

The next useful steps are narrowly specified:

1. Replay both unchanged C++ modes in an assertion-enabled compatible toolchain, recording compiler version, source digest, full exit status, terminal certificates and the split tree. Compare counts only after verifying the underlying arrays and coverage.
2. Export leaf certificates for a small independent checker that verifies substitution, positivity and both-child coverage without trusting the adaptive search heuristic.
3. Test the proven $H \geq -3/25$ substitution in row-b Type T. Report changed coefficient signs and traversal costs, not just the pointwise improvement.
4. Investigate exact factors or face-adapted bases near $p = \boldsymbol{u}$ or $q = \boldsymbol{u}$. A proposed factorization must be algebraically verified before it justifies termination or removes faces.
5. Benchmark sparse coefficient arithmetic against tensor Gram-moment coordinates at matched exact certificates, including transformation cost and integer growth. No advantage is presumed.
6. For approximate posterior computations, retain a separate measure-error bound, support constraints and rational-observable remainder bound. Test observed random-edge and general-offspring consequences only within the independence hypotheses of Section 9.

The correct present conclusion is conditional at the global level: the source supplies an explicit plausible certificate architecture whose mathematical reduction and boundary logic can be audited in detail, and whose scalar portion is independently reproduced. Its full two-variable computational obligation remains unreplayed here. The report adds exact local facts and useful conditional consequences without promoting finite tests into an all-parameter proof.

12 Provenance and reproducibility

12.1 Read scope

All paths below are relative to `C:/Users/mural/OneDrive/Documents/`, unless a different root is stated. The frozen audit ledger supplies absolute paths and SHA256 digests.

Material	Actual read scope
Public M437 source	`OpenAI-722-math/_sources/openai-math/preprints/` followed by the assigned title directory; complete `build/main.tex`, `references.tex`, sections 01–07; complete `verification/checks/certificate-source.txt` and `README.txt`.
Private TR250	`RamFoundationTRs/TR-rev-1-old-tr250/`; directly read finite moment definitions, exact coordinate transfer, concrete polynomial realization and the polynomial Gram construction. Other PDE parts were not relied on for the Potts claim.
Private Volume 16	Complete `RamCalBooks/volumes/volume16/chapters/ch05d-clearance.tex`; particularly Bernstein conversion, subdivision, unresolved zeros, remainder and smoothing budgets.
Private pattern report	`RamPatsCore50-TRs/TR-core-pat-16/TR-core-pat-16.tex`; directly read law/chart definitions, moment contraction and support validity, exact remainder ledger, rank loss and inverse-integrability passages. No patent or priority conclusion is drawn.
Parent artifacts	`OpenAI-722-math/_audit/M437-parent-merge-extension.txt`, `M437-parent-quenched-extension.txt` and `M437-parent-check.json`, read as separately attributed local proof/check evidence.
Build environment	PDF skill and marker script, bundled runtime descriptor and the unchanged local `_tools/compile_report.py`.

Public literature access comprised the Sly and Gu primary abstracts, the Mossel–Sly–Sohn abstract and main tree theorem passages, Leroy's primary PDF and the source's current public README discovery. Failed retrieval of a nonexistent arXiv v4 URL was corrected to the actual v2. No private passage was included in a public query. Search date is October 7, 2026; version labels, rather than search-engine crawl dates, determine publication context.

12.2 Audit outputs and build scope

The author-owned audit script, results and complete scalar coefficient ledger have prefix `_audit/M437-author`. The final source/PDF have stem `TR-M437-Ram-Potts` in `M437-Potts-Reconstruction`. Rendered pages, compiler logs and extraction diagnostics reside under `_build/TR-M437-Ram-Potts/round-1` and `round-2`. The unchanged helper performs three TeX passes per formatting round. Native editor/compiler availability is separately recorded in the build audit; no native success is inferred from the installed-MiKTeX export.

The final manifest records inputs, source read scope, public URLs, audit artifacts, final outputs and all author-owned changed paths. Its own digest is reported separately to avoid a self-referential hash. The freeze occurs only after the final PDF has been rendered and inspected. Nothing in the shared catalogue, queue, progress record, helper directory or private originals is edited by this report.

References

[1] OpenAI. *The Reconstruction Threshold for the Ferromagnetic Four-State Potts Model*. October 5, 2026. Pinned source snapshot `adc7f1241b42e322a6451854ab7e4b4c146bf78a`; source Sections 1–7. Public repository: `https://github.com/openai/math/tree/adc7f1241b42e322a6451854ab7e4b4c146bf78a`. The public claim is source-reported, not independently fully certified here.

[2] A. Sly. *Reconstruction of symmetric Potts Models*, primary preprint, 2008; published as *Reconstruction for the Potts model*, Annals of Probability 39 (2011), 1365–1406. `https://arxiv.org/abs/0811.1208`; `https://doi.org/10.1214/10-AOP584`.

[3] E. Mossel, A. Sly and Y. Sohn. *Exact Phase Transitions for Stochastic Block Models and Reconstruction on Trees*. Revised February 22, 2025, v2, Theorems 1.12–1.13. Annals of Probability 53 (2025), 967–1018. `https://arxiv.org/abs/2212.03362`; `https://arxiv.org/html/2212.03362v2`; `https://doi.org/10.1214/24-AOP1723`.

[4] Y. Gu. *Exact reconstruction thresholds on hypertrees over a symmetric binary alphabet*. June 19, 2026, v1; primary abstract inspected. `https://arxiv.org/abs/2606.21699`.

[5] R. Leroy. *Convergence under Subdivision and Complexity of Polynomial Minimization in the Simplicial Bernstein Basis*. Reliable Computing 17 (2012), 11–21. `https://www.reliable-computing.org/reliable-computing-17-pp-011-021.pdf`.

[6] M. Subbarao. *Compatible Variational Ram Panels*, TR250, foundation review September 20, 2026. Confidential author-supplied local source. Finite moment coordinates, compatible coordinate transfer and concrete polynomial construction; precise local path in the audit ledger.

[7] M. Subbarao. *Ram Calculus*, Volume 16, chapter fragment `ch05d-clearance.tex`. Confidential author-supplied local source. Continuous constraints, exact Bernstein enclosures, subdivision and remainder qualifications.

[8] M. Subbarao. *Joint-Moment Bayesian Neural Inference and Inverse Uncertainty Propagation*, TR-core-pat-16, October 1, 2026. Confidential author-supplied local source. Sections on joint laws, covariance, support validity, remainder bounds and rank loss.

[9] Parent audit for M437. *Local extension and certificate boundary*, *Quenched and survival-conditioned extension*, October 7, 2026, and exact-check receipt. Local files `M437-parent-merge-extension.txt`, `M437-parent-quenched-extension.txt` and `M437-parent-check.json`; no full C++ replay asserted.